

On-Chain Parametric Cost Insurance for Underserved Latin-American Digital Workers

Record of Closed Iterations and Request for Methodological Guidance

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Purpose

This letter is the entry-point document for the supervisor’s review of the analytics workstream. It records the five iterations that have been closed or parked to date, states the proposition motivating the iteration design, and asks five methodological questions in priority order whose answers we expect to bind the next iteration’s pre-registration. A 37-page technical companion at [docs/supervisor-review/ 2026-05-06-empirical-record-letter.pdf](#) provides the per-iteration estimation detail, the verdict-mapping decision tree, and a sample artifact for any question the supervisor wishes to drill into. The implementation venue is on-chain (Panoptic on Uniswap v4) but the empirical work being reviewed here is off-chain econometrics; the on-chain primer and the full methodology are deferred to the companion.

§1. The empirical question and the iteration roster

The project’s economic motivation is best stated as a financial-intermediation gap for an identifiable underserved population rather than as an abstract claim about aggregate inequality. The population the iteration roster currently centres on is Latin-American digital workers paid in local currency (Colombian pesos, in the dev-AI Stage-1 iteration) who carry a recurring USD-denominated cost burden through their professional inputs — subscription AI services, application programming interface usage fees, cloud compute, and adjacent software-as-a-service spend. These workers are exposed to FX-pass-through cost shocks on their monthly USD spend that no Colombian retail-bank product currently hedges. Conventional FX-derivative venues are inaccessible: minimum notional sizes, counterparty credit standing, and intermediating-broker requirements are calibrated for corporate treasury clients, not for an individual digital worker carrying a hundred-to-low-thousands USD monthly cost line. The intermediation gap is the operative phenomenon; aggregate distributional inequality is, under this reading, the cumulative consequence of many such uninsured intermediation gaps rather than the framing primitive.

The wage-to-capital boundary remains a documented empirical phenomenon and is preserved as the analytical object of the iteration roster. [Mendieta-Muñoz \(2017\)](#) documents premature dein-

dustrialization in Colombia and the corresponding reallocation of labor away from productivity-rising sectors; [Rodrik \(2016\)](#) establishes the cross-country pattern; [McMillan and Rodrik \(2011\)](#) formalises the structural-change-and-productivity decomposition that underlies the pattern. Together these references establish that for a substantial subset of developing-economy populations, the path from wage income into productive-capital ownership has become structurally narrower over the past three decades, and that the populations on the wrong side of that narrowing are exactly those that an absent intermediation layer fails to insure. The project’s empirical question is which macro risks X are quantitatively responsible for the recurring uninsured cost shocks that keep specific target populations on the wrong side of this boundary.

The unit of work is a triple (Y, M, X) . Y is the welfare-relevant outcome variable on which a target population’s exposure to the wage-to-capital transition is measured (operationalised per population: the Section-J information-and-communication employment share for young Colombian workers, in the dev-AI Stage-1 iteration). X is the macro risk hypothesised to block the transition for that population, identified empirically and entered as a lag-distributed right-hand-side variable under a pre-pinned lag structure. M is the settlement geometry of the candidate hedging instrument on the on-chain substrate; it is declared out of scope for empirical validation and deferred to Stage 2 of each iteration. The validation strategy estimates a composite slope $\hat{\beta}_{\text{composite}}$ of Y on lagged X , with sign expectation, lag structure, primary specification, and verdict thresholds pre-registered before the panel is touched. The methodology is a pre-registered composite- β OLS+HAC validation; full detail in the companion.

As of the date of this letter, the project has run five iterations and *all five are closed or parked*: no Stage-1 work is in progress. This letter therefore presents a complete log rather than a snapshot.

§2. Five iterations: a compact record

The dev-AI Stage-1 Section-M sensitivity arm result deserves a sentence of cross-iteration framing rather than only a per-row caveat. Read jointly with the Pair D PASS, the pattern — Section J narrow-ICT sign-flipped negative, Section M (professional/scientific/technical/administrative services) strongly positive on the same FX-lag panel and same monthly cadence — suggests that the Pair D PASS on the broad Section-G–T cut was carried by the Section-M-style sub-aggregate rather than by Section-J ICT-narrow tech offshoring. The Pair D headline numbers stand verbatim under the original spec; what has sharpened is the interpretation of the operative sub-aggregate. The compositional reframe is recorded in the companion (§6.2) and is one of the inputs to the cross-iteration multiple-comparison question in §4.5.

The cross-iteration multiple-comparison structure is itself an open methodological concern. Each iteration was pre-registered individually; the project-level decisions that produced the roster (which (Y, X) pairs to spec-pin, which to drop after EXIT, which to deepen after PASS) are not bound by an external pre-registration rule. The numerical mass at the two tails across the five tests is asymmetric in a way that matters for any FWER-style correction: Pair D’s one-sided $p \approx 1.46 \times 10^{-8}$ is overwhelmingly significant against any reasonable [Benjamini and Hochberg \(1995\)](#)- or [Romano and Wolf \(2005\)](#)-style correction across five iteration-level tests; the FX-vol-

Population	(Y, X) pair	Verdict	Headline number	One-line caveat
Colombian young workers (14-28), broad services	Sections G-T youth share (logit) on $\log(\text{COP}/\text{USD})_{t-k}$, $k \in \{6, 9, 12\}$	PASS (2026-04-28)	$\hat{\beta}_{\text{composite}} = +0.13670985$; HAC SE 0.0247; $p \approx 1.46 \times 10^{-8}$; $N = 134$	Hedges correlation, not causal channel; effect concentrated at six-month lag; sub-aggregate may be Section M, not Section J (see below).
Colombian young workers (14-28), Section J ICT (dev-AI Stage-1)	Section-J youth share (logit) on $\log(\text{COP}/\text{USD})_{t-k}$, $k \in \{6, 9, 12\}$	FAIL, sign-flipped (2026-05-06)	$\hat{\beta}_{\text{composite}} = -0.146$; HAC SE 0.0847; $t = -1.726$; $N = 134$	High-information FAIL: pre-registered $H_1 : \beta > 0$ rejected at the negative sign. Pre-specified Section-M sensitivity arm returned $\hat{\beta} = +0.455$, $p = 1.13 \times 10^{-6}$ (not pre-registered as primary; candidate next iteration).
Colombian corporates and remittance recipients	Weekly TRM realised volatility on signed AR(1) CPI surprise	FAIL on weekly primary (2026-04-19)	$\hat{\beta}_{\text{CPI}} = -0.000685$; HAC(4) 90 % interval $[-0.0036, +0.0023]$; $N = 947$ weekly	A1 monthly cadence and A4 release-day-excluded sensitivities both pre-registered, both with 90% intervals strictly above zero; A4 lower bound 0.0005 is small in absolute magnitude. T1 exogeneity rejected ($F = 15.12$): predictive, not structural.
Colombian household remittance recipients	Weekly TRM realised vol on candidate on-chain remittance-flow proxy (cCOP/COPm)	EXIT_NON_REMITTANCE (2026-04-24)	None: apparatuses; kill-criterion exit at panel-construction stage	0/30 peak-day flows fingerprinted as remittance; $\sim 87\%$ of surveyed flow decomposed into TRM arbitrage, treasury roundtripping, campaigns, and bot activity. Kill criterion k1 fired (intent non-remittance dominant).
LATAM AI-policy-shock-exposed (downstream)	Bittensor SN18 (Cortex.t) event-study apparatus on policy-event dates	PARKED (2026-04-27)	None: apparatus complete, data ingest deferred	Slow-lane apparatus parked to preserve intellectual capital after pivot to fast-lane simple- β track (Pair D, dev-AI Stage-1). Re-activation conditions documented in parking memory.

Table 1: The five iterations. All numerical claims trace to §6 of the companion document. Pair D, dev-AI Stage-1, and FX-vol-CPI-surprise are econometric estimations; Phase-A.0 remittance and P1 Bittensor SN18 are non-estimation closures. $N_{\min} = 75$, $\text{POWER}_{\min} = 0.80$, $\text{MDES}_{\text{SD}} = 0.40$ are valid in iteration pre-registration and finishing. Invariantes applied uniformly across the master 2026

CPI sensitivity arms (A4 lower bound 0.0005) are not. The full treatment is the subject of §4.5.

§3. The hedge-target proposition

Lemma 1 (Inequality-Differential Hedge Target). *Let $R_a(t)$ denote the return process of an asset basket representative of capital-holding households' wealth, and let $R_c(t)$ denote the return process of a consumption basket representative of working-class households' purchasing power. Define the inequality-differential process*

$$Y_{\text{inequality}}(t) = R_a(t) - R_c(t).$$

A hedging instrument settled on $Y_{\text{inequality}}(t)$ is welfare-relevant for working-class wealth in a way that no instrument settled on $R_a(t)$ alone or $R_c(t)$ alone is, because the differential directly indexes the relative-wealth gap that determines the working-class household's purchasing-power decline relative to capital-holding households.

Lemma 1 is stated as a proposition motivating the iteration design, not as a proven theorem; the calibration of the joint-leg structure remains Stage-2 work and is the subject of §4.3 below. The lemma is read here under the parametric-cost-insurance underwriting standard (Carter et al., 2017; Mahul and Stutley, 2010; Clarke, 2016) rather than under generic option-pricing standards: the working-class household holds the policy and the differential serves as the welfare-relevant index variable on which payouts settle.

§4. Five methodological questions

§4.1 Identification strategy

The Stage-1 specification is OLS on lagged log-FX with HAC inference. $\hat{\beta}_{\text{composite}}$ is a predictive-regression coefficient and we do not treat it as an impulse-response. The Pair D closure explicitly hedges the correlation rather than asserting a BPO causal channel; the FX-vol-CPI iteration's T1 exogeneity test rejected at $F = 15.12$, confirming that exercise is also predictive rather than structural. *Question.* Should subsequent iterations move from a predictive-regression posture to a causally-identified one — and if so, what identification standard does the supervisor require before declaring a Stage-1 PASS sufficient for Stage-2 dispatch? Candidate strategies under consideration are (i) instrumental variables, with central-bank policy-rate decisions, oil-price shocks, or US-side AI-API price changes as candidate instruments (per Hansen, 2022, Chs. 6, 12); (ii) event-study formulations leveraging the apparatus already developed for Phase-A.0 remittance and parked P1 Bittensor SN18; (iii) structural VAR per Kilian and Lütkepohl (2017), for which we do not currently have a defensible identification scheme. *Action we will take.* We will adopt the supervisor's preferred standard as the binding identification specification for the next iteration's pre-registration.

§4.2 Convex-payoff sufficiency under the insurance-underwriting standard

Stage-1 estimates a mean slope. Mean- $\hat{\beta}$ identification is necessary but not sufficient for pricing payoffs that transmit through the upper or lower tail of the conditional distribution rather than through the conditional mean: this is the basis-risk question of [Clarke \(2016\)](#). *Question.* What empirical evidence does the supervisor require before this $\hat{\beta}$ -validation is sufficient to underwrite a parametric cost-insurance product on the validated X ? Specific sub-questions: (i) is variance-amplification evidence at X -shock dates sufficient (e.g. a GARCH- X specification with the validated X entering the conditional-variance equation); (ii) is full quantile-regression coverage of $Y \mid X_{t-k}$ required so that the basis-risk envelope of [Clarke \(2016\)](#) is directly observable; (iii) is extreme-value-theory peaks-over-threshold tail-index estimation ([Embrechts et al., 1997](#)) required to bound the catastrophic-loss probability that [Carter et al. \(2017\)](#) and [Mahul and Stutley \(2010\)](#) treat as a first-order underwriting input; (iv) should the dynamic conditional-correlation MGARCH of [Engle \(2002\)](#) be the standard joint-distribution model for the differential's two legs. *Action we will take.* We will adopt the supervisor's chosen subset as the Stage-1 exit criterion for Stage-2 dispatch.

§4.3 Multi- Y differential calibration

Lemma 1's differential $Y_{\text{inequality}} = R_a - R_c$ requires constructing each leg as a panel of observable component variables. Closed iterations estimate single- Y , single- X responses; the joint-leg structure has not been calibrated and the within-iteration thresholds were derived under a single- Y assumption. *Question.* How should the joint-leg structure be calibrated? Specifically: (i) joint-leg correlation — constant-correlation GARCH, the dynamic conditional-correlation specification of [Engle \(2002\)](#), or a regime-switching extension; (ii) within-leg weighting — fixed welfare-anchored weights, principal-components weights, or panel-estimated weights; (iii) regime-switching — Markov-switching mean-and-variance to accommodate the documented regime mix (post-2014 oil collapse, COVID, Fed-tightening) noted in the Pair D closure, or regime indicators as exogenous covariates only; (iv) familywise error under the multi- Y panel — Bonferroni-Holm versus false-discovery-rate, given that within-leg components are expected to be highly pairwise correlated and Bonferroni is therefore likely conservative. *Action we will take.* We will adopt the supervisor's chosen joint specification as the prerequisite for promoting Lemma 1 from a motivating proposition to a calibrated estimation target.

§4.4 Anti-fishing protocol defensibility

The within-iteration anti-fishing invariants are $N_{\min} = 75$, $\text{POWER}_{\min} = 0.80$, and $\text{MDES}_{\text{SD}} = 0.40$ standard-deviation units of Y . These three thresholds were inherited from the Phase-A.0 Rev-5.3.1 specification and have not been re-derived under each subsequent iteration's actual N and observed first-order autocorrelation structure of the residuals. *Question.* Are these three thresholds defensible from the supervisor's standpoint, and what additional pre-registration discipline does the supervisor require? Specifically: (i) should the project register pre-analysis plans in a public registry in the spirit of [Olken \(2015\)](#) on pre-analysis plans in development economics and [Christensen and Miguel \(2018\)](#) on transparency and reproducibility in economics; (ii) should

the project adopt the AEA RCT-registry analogue for observational pre-registration (current discipline pins spec-text by sha256 hash inside the project repository, auditable internally but not externally registered); (iii) should the three thresholds be re-derived per iteration, taking each iteration's observed N and HAC-corrected effective sample size as inputs to the power calculation (in the spirit of Hansen, 2022, §6.18). *Action we will take.* We will adopt the supervisor's preferred pre-registration infrastructure for subsequent iterations, including whether the inherited threshold triple should remain in force or be replaced.

§4.5 Cross-iteration FWER and iteration-selection bias

The five-iteration roster is closed; the cross-iteration structure is not pre-registered. The mass at the two tails of the iteration-result distribution is asymmetric: Pair D's one-sided $p \approx 1.46 \times 10^{-8}$ survives any reasonable Bonferroni correction across five iteration-level tests; the FX-vol-CPI-surprise sensitivity arms (A1 monthly: 90% interval $[+0.0057, +0.0246]$; A4 release-day-excluded: $[+0.0005, +0.0062]$) are differentially robust to such a correction; the dev-AI Stage-1 primary delivered $\hat{\beta} = -0.146$ at $t = -1.726$, recorded as a *sign-flipped* FAIL with the realised sign at the direction of the alternative rather than merely insignificant. *Question.* What FWER or false-discovery-rate correction regime should be applied across the iteration history given the directional asymmetry, and how should sensitivity-positive findings such as the FX-vol A1 and A4 results be weighed under such a correction? In particular, should a sign-flipped FAIL at conventional significance be counted symmetrically with the PASS at the positive tail, or should the directional pre-registration on each iteration shift the mass allocation? A second sub-question is methodologically prior: iteration selection itself — which (Y, X) pair to specify, which to drop, which to deepen — is a multiple-testing structure that within-iteration pre-registration does not address. Candidate mitigations include pre-registered iteration-budget caps committed at project initiation, an iteration-level alpha-spending function in the spirit of group sequential designs, and pre-registered iteration-allocation rules that do not depend on prior verdicts. A third, sharpened sub-question: is the Pair D PASS one test, or one test conditional on a sub-aggregate selection? The dev-AI Stage-1 Section-M sensitivity arm's positive estimate now suggests that the sub-aggregate concentrating the broad-services signal is Section M rather than Section J; if the Pair D PASS is read with this in mind, the multiple-comparison ledger must additionally account for an implicit search across sub-aggregates of the broad-services panel (the selection-on-observables frame of Heckman, 1979, is one candidate reference). *Action we will take.* We will adopt the supervisor's preferred FWER/FDR regime, the supervisor's preferred iteration-selection mitigation, and the supervisor's reading of the Pair D-conditional-on-sub-aggregate question as binding constraints on the next iteration's design.

Closing and pointer to the companion

Five iterations are closed or parked; no Stage-1 work is in progress. The next iteration's pre-registration is held open pending the supervisor's response to the five questions above. The 37-page technical companion at `docs/supervisor-review/2026-05-06-empirical-record-letter.pdf` contains the full per-iteration record (companion §6, nine-field schema per iteration), the on-chain

settlement primer (§4), the methodology in full (§5, including the verdict-mapping decision tree), and four appendices: Glossary (A), Cited Literature (B), Pair D Verdict Header sample artifact (C), and Power-Calculation Table (D). The supervisor is invited to drill into any iteration's per-iteration block in companion §6 for the specification chain, the sha256 pre-registration governance, and the independent-review caveats inherited from the closure memoranda.

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